What is the best small AI model for inpainting-based synthetic image detection?

Final presentation

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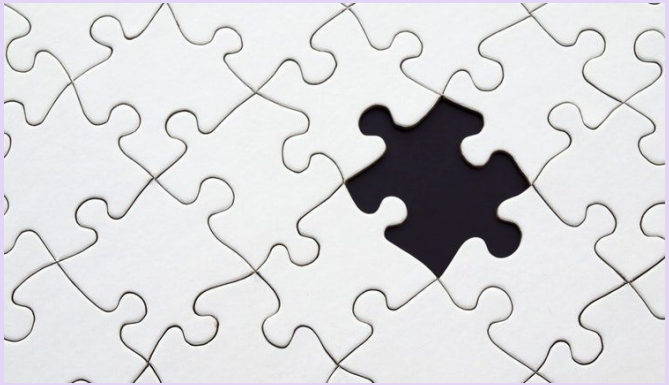
Inpainting

restoring missing or damaged regions of an image using information from the surrounding pixels



The Gap

Heterogeneous experimental settings, difficult to compare selected small models across performance, generalization, and computational efficiency.



The need for a **unified benchmarking framework**

Research Question

RQ1 What are the categories or models and different architectures used to detect / localize inpainting?

RQ2 Which evaluation criteria are appropriate for comparing small AI models for inpainting detection and localization?

RQ3 How do selected models compare under a standardized evaluation protocol?

RQ4 What trade-offs exist between detection / localization performance and computational cost among selected models?

Criteria to choose the models

1. **Different families:** one model per approach (CNN, state space, transformer, reconstruction).
2. **Efficiency:** lightweight or low-compute design, since our question is about small models.
3. **Both tasks, or adaptable:** detection and localization.
4. **Relevant:** built for image manipulation, ideally for generated edits.

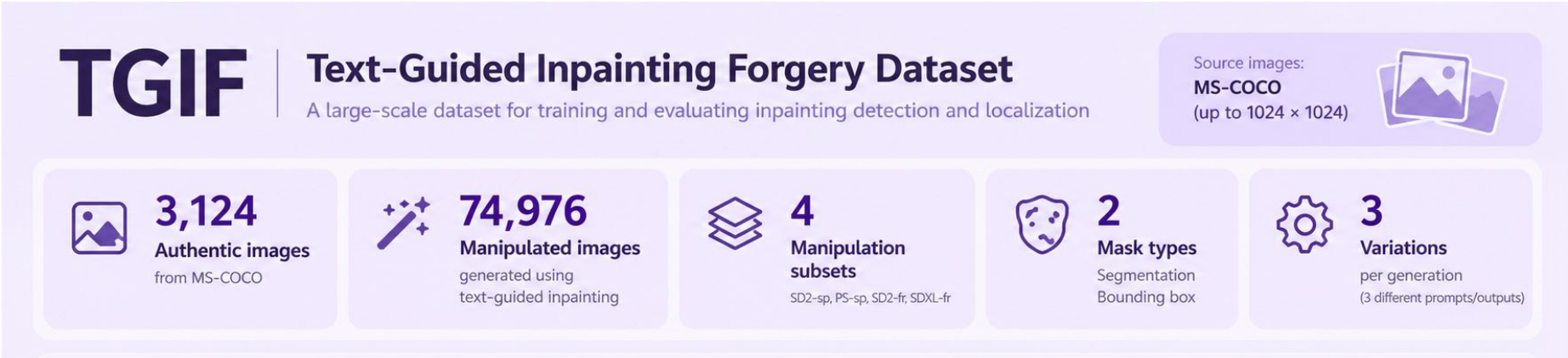
Detection and localization architectures

CNN-based	PSCC-Net
Transformer-based	TruFora
Reconstruction-based	AEROBLADE
State Space Models	ForMa

Dataset: TGIF

Why is it relevant ?

- Large-scale dataset
- Diversity of generative methods
- Relevance to our research question
- Recent and established benchmark



Dataset: TGIF

Why is it relevant ?

It is compliant with the **F.A.I.R** principles

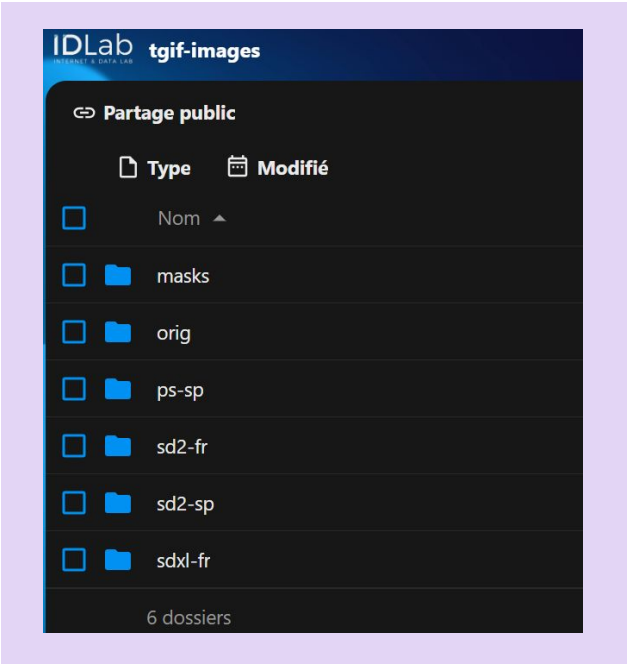
F Findable: Data and metadata must be easy to locate

A Accessible: Once data is found, users must know how to retrieve it.

I Interoperable: Data must integrate and work well with other datasets, storage systems, or analytical tools.

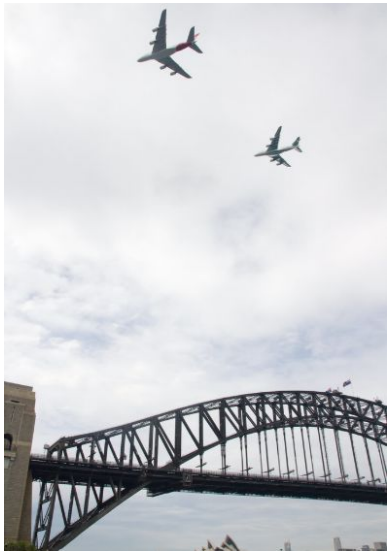
R Reusable: Data must be documented well enough to be replicated and used in new studies

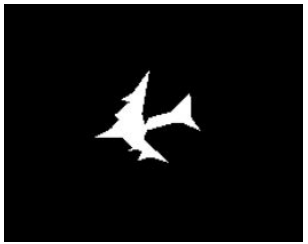
More about the TGIF Dataset

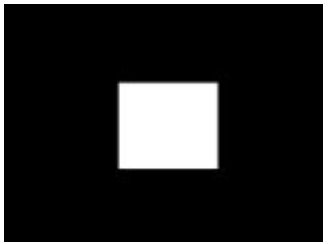


We focus on splicing and not fully regenerated images: either with photoshop and adobe firefly or stable diffusion

Inpainted Input Image







Masks: Localization Ground Truth

Evaluation Metrics

Detection

- Accuracy
- F1 score
- ROC-AUC

Localization

- IoU
- Dice Score
- Pixel-wise Recall
- Pixel-wise Precision

Efficiency Metrics : Compute power, Model Size, trade-off

Evaluation Metrics

Visualizing Localisation Metrics



Ground Truth Mask (G)

$|G| = 100$ pixels



Predicted Mask (P)

$|P| = 120$ pixels



Intersection ($P \cap G$)

$|P \cap G| = 80$ pixels



Union ($P \cup G$)

$|P \cup G| = 140$ pixels

IoU

$$\frac{80}{140} = 0.571$$

(57.1%)

Dice Score

$$\frac{2 \times 80}{100 + 120} = 0.727$$

(72.7 %)

General Results: DETECTION

Combining all 4,116 images across 13 categories and both generators (PS and SD)

	TrueFor	ForMa	PSCC-Net	AEROBLADE
Accuracy	0.95	0.77	0.66	0.57
precision	0.98	0.93	0.86	0.92
recall	0.96	0.82	0.69	0.59
F1	0.97	0.87	0.76	0.70
ROC-AUC	0.97	0.48	0.61	0.51

General Results: Localization

Combining all 4,116 images across 13 categories and both generators (PS and SD)

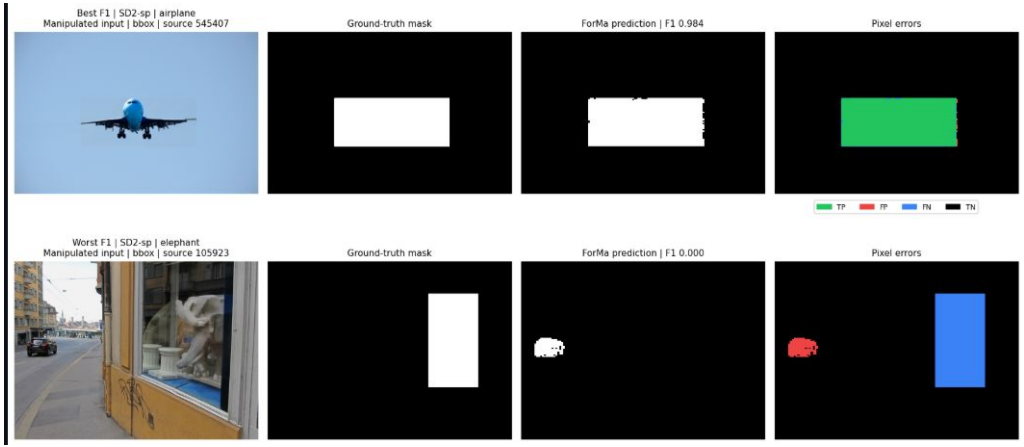
	TrueFor	PSCC-Net	ForMa	AEROBLADE
IoU	0.72	0.33	0.21	0.07
Dice	0.87	0.41	0.29	0.13
precision	0.93	0.46	0.41	0.08
recall	0.82	0.48	0.30	0.70

General Results: Time

Model	Params	GFLOPs / image	Latency / image (GPU, median)	Throughput	Peak RAM	Peak GPU mem	Est. full TGIF test (4,459 img)
AEROBLADE	98.4 M	3,841	891 ms (mean 833)	1.20 img/s	1,160 MiB	1,351 MiB	**≈ 62 min**
TruFor	68.7 M	1,255	872 ms (mean 836)	1.20 img/s	992 MiB	2,395 MiB	**≈ 62 min**
ForMa	37.3 M		11,628 ms (mean 11,659)	0.09 img/s	1679 MiB	446 MiB	**≈ 14.4 h**
PSCC-Net	3.7 M	542	856 ms (mean 897)	1.12 img/s	2,796 MiB	2,527 MiB	**≈ 67 min**

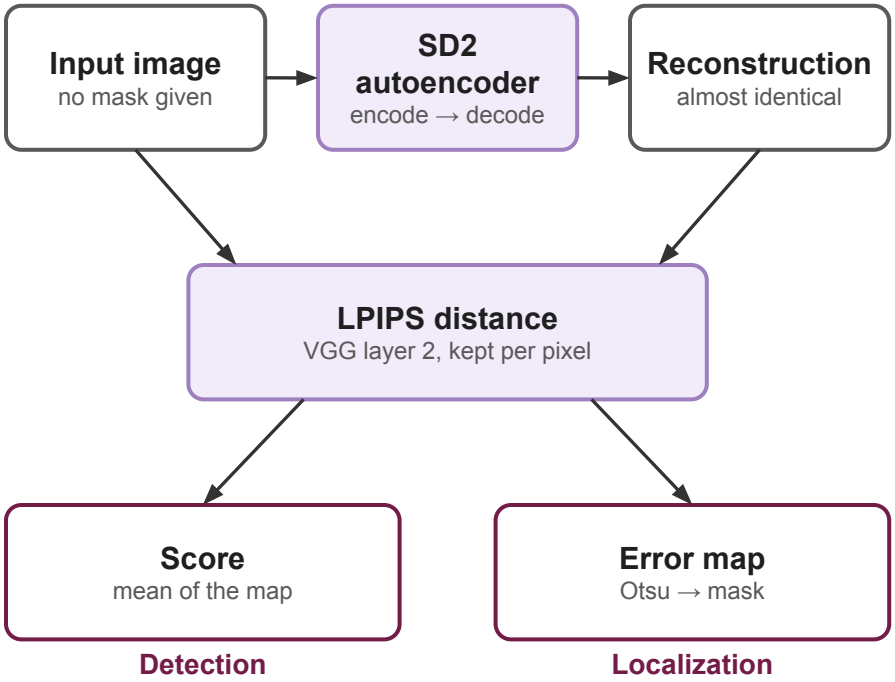
General Observed Patterns

- Localization is more sensitive to changes in inpatining method (PS, SD2)
- Localization and detection is generally more difficult on small objects than big items



AEROBLADE: Reconstruction-Based Detection and Localization

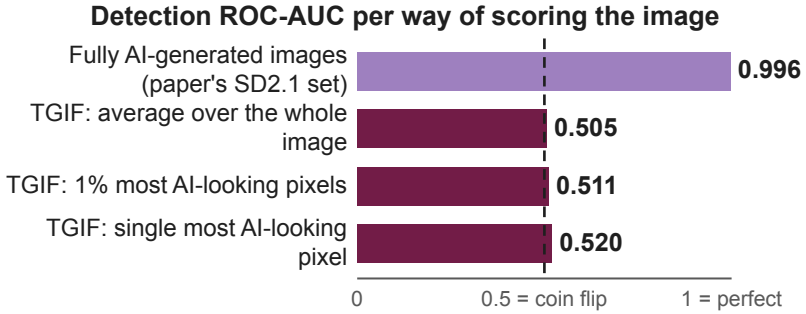
- Training-free: 1 autoencoder pass
- Low error = generated
- Mean of the map → detection
- Map itself → localization (our addition)
- SD2 autoencoder, ≈84M parameters



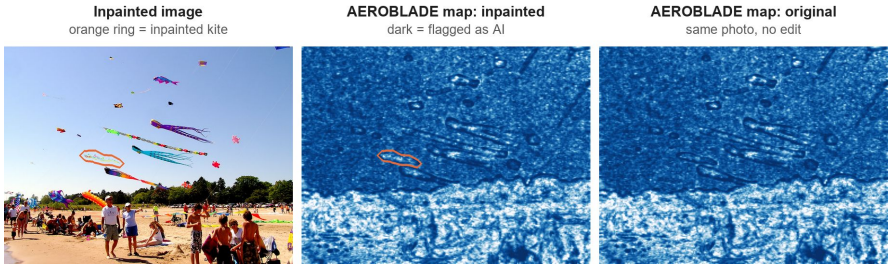
All 4,459 TGIF test images · 512×512 · laptop CPU, 18.4 s/image

Results: detection at chance, localization weak

Detection ROC-AUC 0.51 (0.5 = coin flip)



Localization IoU 0.07 · Dice 0.13 (1 = perfect)



AI-likeness: **real sky 0.82** · **inpainted kite 0.60**

Why?

- Every way of scoring the image is at chance
- Even the single most AI-looking pixel: 0.52
- Inpainted pixels don't look more AI than real ones (see the kite)

Why?

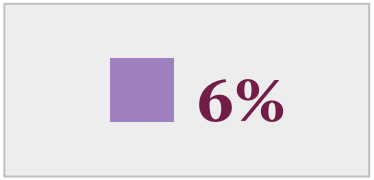
- No averaging here: one value per pixel
- The real sky is flagged, even in the untouched original
- Smooth areas rebuild easily → look AI: the map shows content, not the edit

Fully AI-generated: paper's own SD2.1 images vs its 1,000 real photos (our code). TGIF: all 4,459 test images. AI-likeness: map value (0–1) averaged over the sky (top third) and over the kite.

Why AEROBLADE fails on TGIF: 3 reasons

Our code is correct: same AP as the paper on the paper's own data (≥ 0.99).

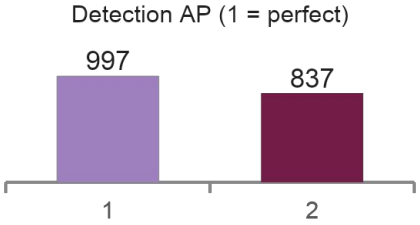
1 The edit is tiny



grey = real photo · purple = inpainted part

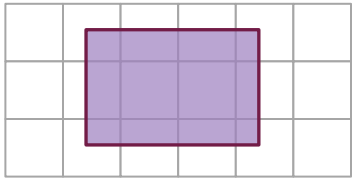
Only 6% of the image is AI-made, and it doesn't look more "AI" than the rest.

2 Resizing blurs the trace



Paper's own Midjourney images: AP 0.997 → 0.837 after our 512×512 resize.

3 The patch is off the 8×8 grid



squares = autoencoder blocks · purple = pasted edit

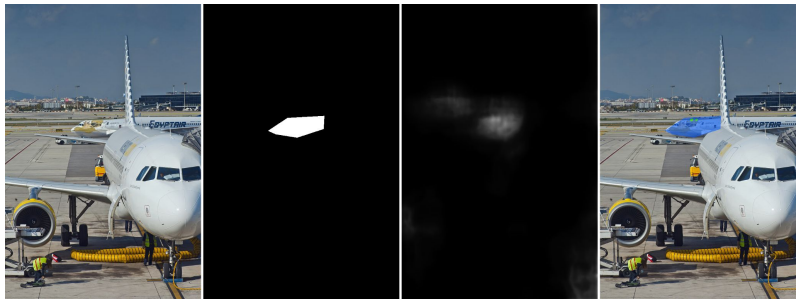
TGIF pastes the edit between the autoencoder's blocks.
Re-aligned*: ROC-AUC 0.66 → 0.82.

* uses TGIF's recorded paste position (upper bound) · 40 images at native size · pixel ROC-AUC 0.5 = chance

PSCC-Net - localization threshold

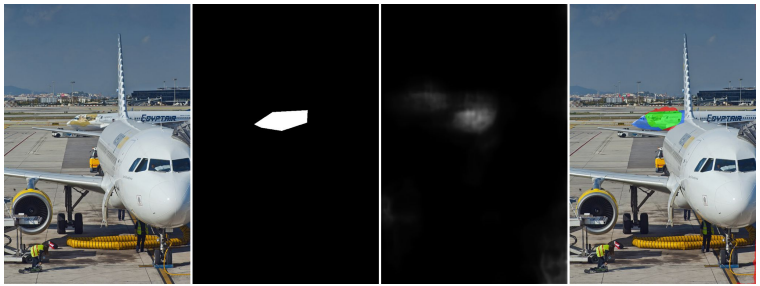
0.5

IoU	0.21
Dice	0.27
Precision	**0.45**
Recall	**0.24**



0.2

IoU	0.33
Dice	0.41
Precision	**0.46**
Recall	**0.48**



Conclusion: A lower threshold doubles recall without losing precision: PSCC-Net detects forgeries but is under-confident.

PSCC-Net - gated vs not gated (threshold 0.2)

not gated

IoU	0.33
Dice	0.41
Precision	0.46
Recall	0.48

gated

IoU	0.23
Dice	0.29
Precision	0.33
Recall	0.34

Conclusion: Detection is the bottleneck of PSSC-Net. The two heads work better independently.

PSSC-Net - other variables

Detection

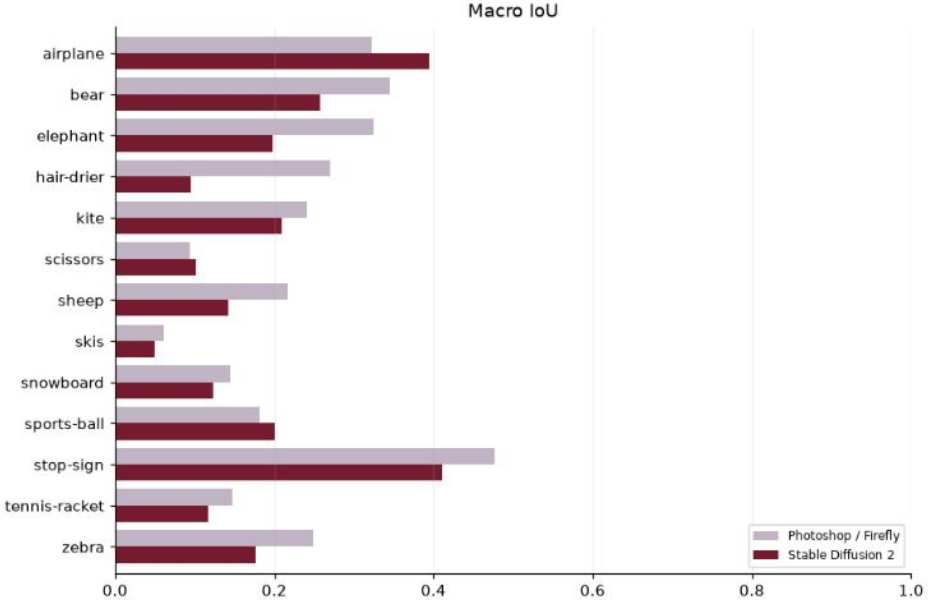
- results independent on generator method, mask type, and variant

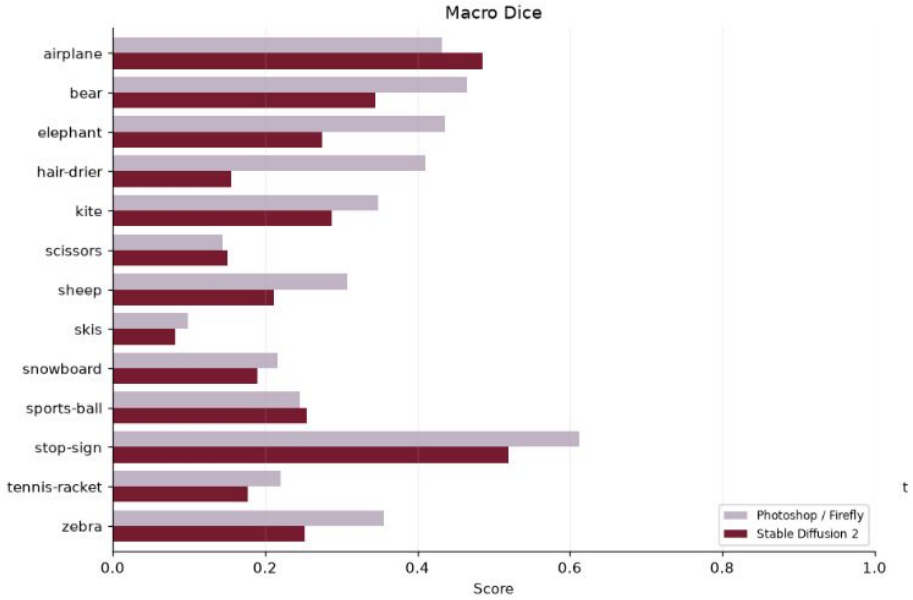
Localization

- dependent on generation method
  - PS (IoU=0.30, Dice=0.38)
  - SD2 (IoU=0.12, Dice=0.15)
- huge range for categories

category	iou_mean	dice_mean	pix_precision_mean	pix_recall_mean
skis	0.0515	0.0703	0.1539	0.0570
snowboard	0.0605	0.0835	0.1991	0.1039
tennis racket	0.0852	0.1165	0.2305	0.1344
scissors	0.1316	0.1723	0.3441	0.1495
sports ball	0.1352	0.1776	0.2859	0.1544
kite	0.1549	0.1998	0.3903	0.1907
airplane	0.2069	0.2747	0.4891	0.2676
hair drier	0.2336	0.2998	0.5810	0.2349
elephant	0.2937	0.3783	0.6756	0.3050
stop sign	0.3066	0.3786	0.5799	0.3825
sheep	0.3204	0.4023	0.6234	0.3568
bear	0.3688	0.4675	0.7270	0.3882
zebra	0.4299	0.5348	0.7901	0.4700

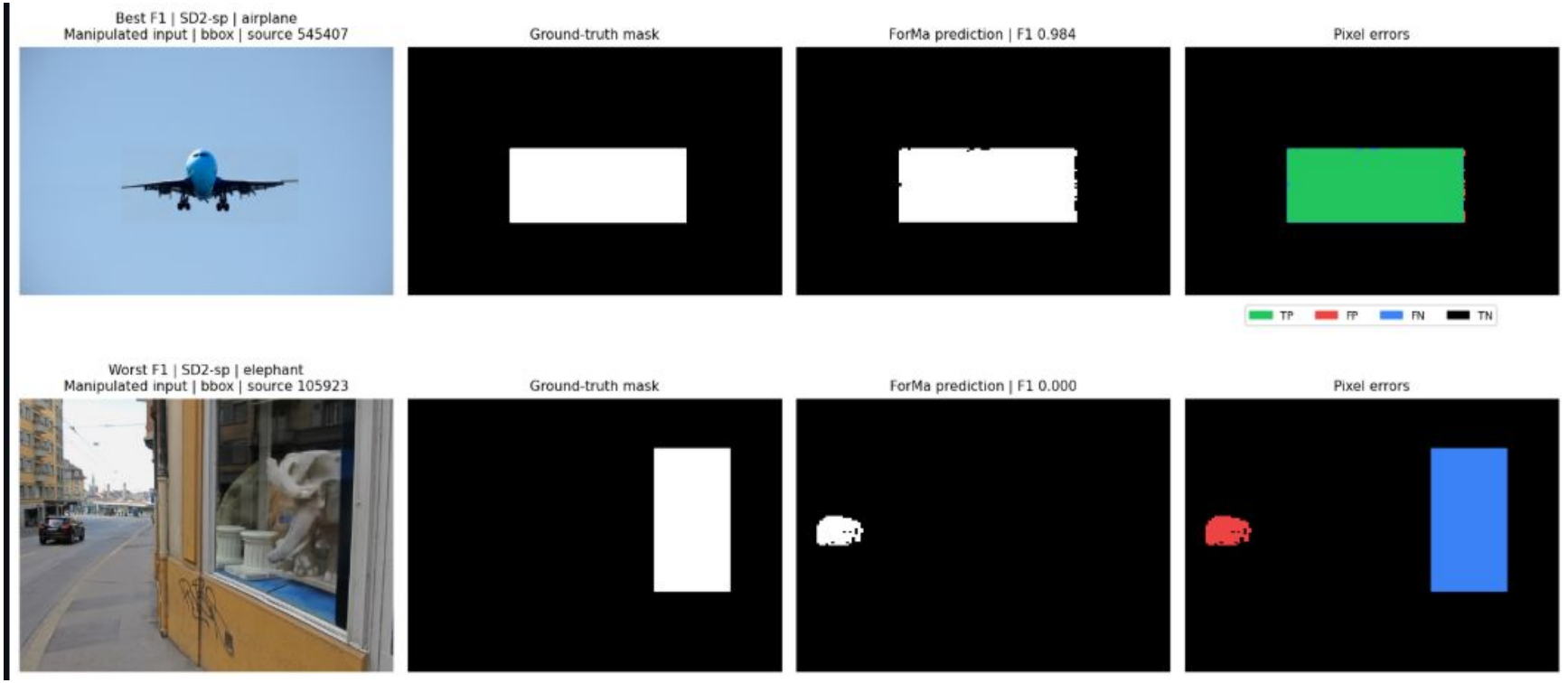
ForMa Results Interpretation



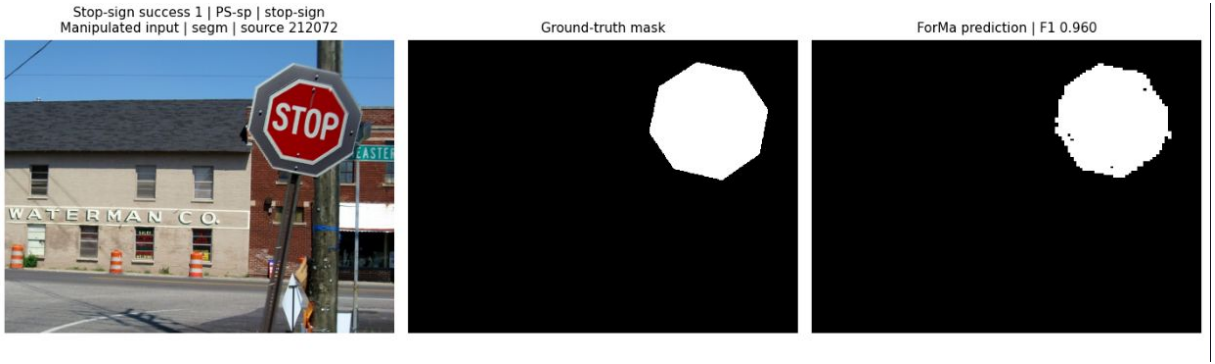


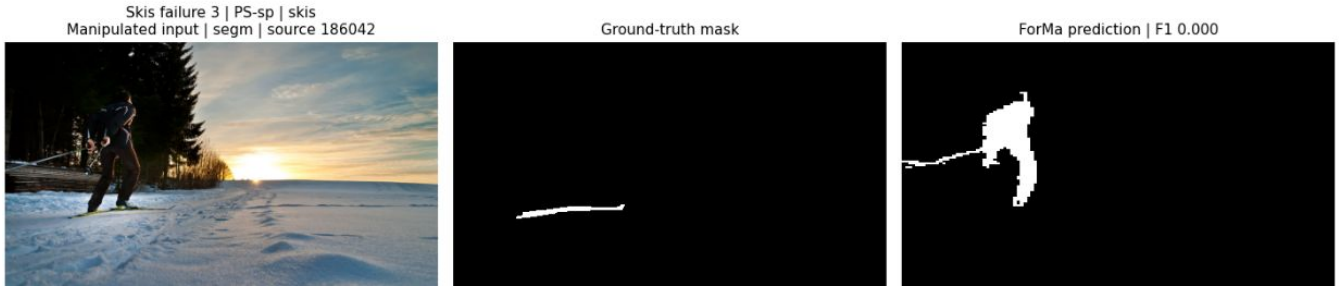
- We notice better results for Photoshop compared to stable diffusion

ForMa


Best F1 | SD2-sp | airplane
Manipulated input | bbox | source 545407
Ground-truth mask
ForMa prediction | F1 0.984
Pixel errors
TP
FP
FN
TN
Worst F1 | SD2-sp | elephant
Manipulated input | bbox | source 105923
Ground-truth mask
ForMa prediction | F1 0.000
Pixel errors

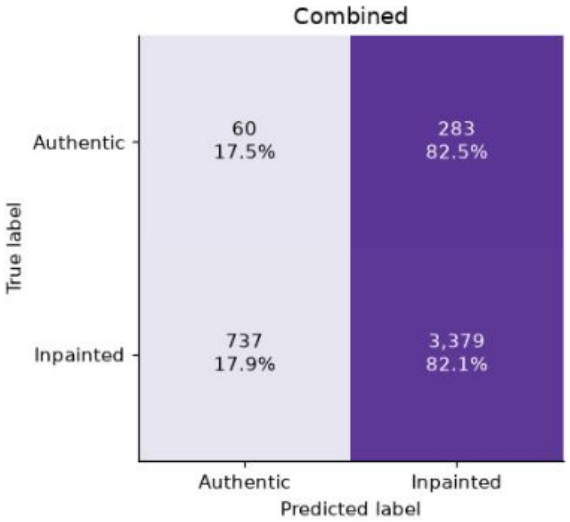
ForMa

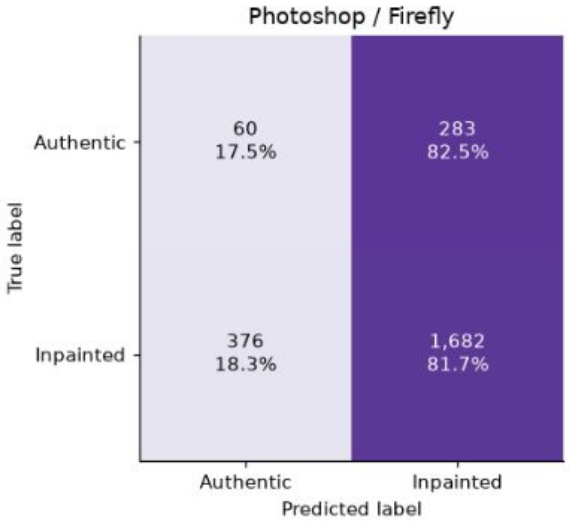


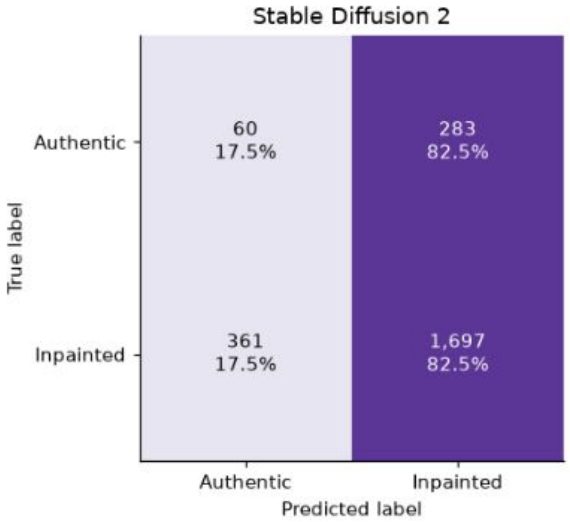


One possible explanation is the geometry of the manipulated regions: compact regions may be easier to localize, while thin, elongated regions are more sensitive to boundary errors, reducing IoU and Dice. Differences in edited-area size, background complexity, may also contribute.

ForMa



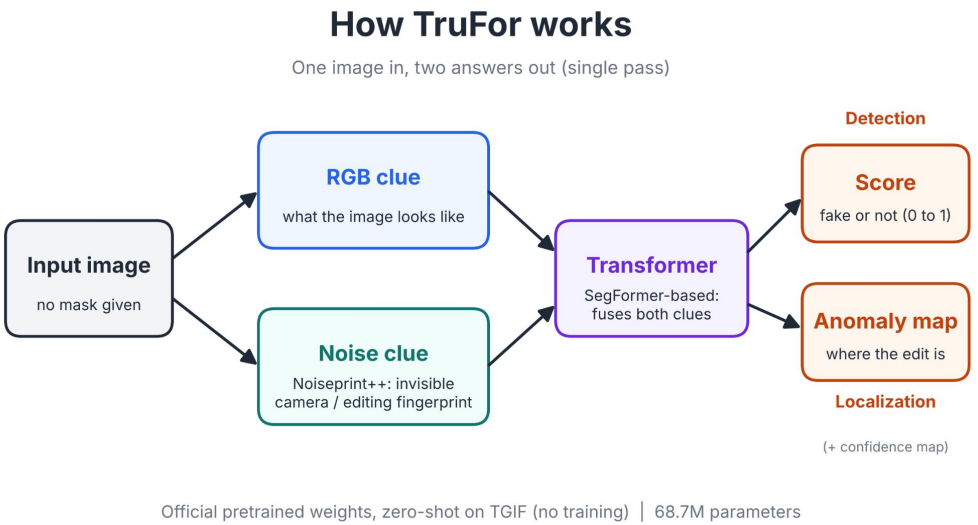




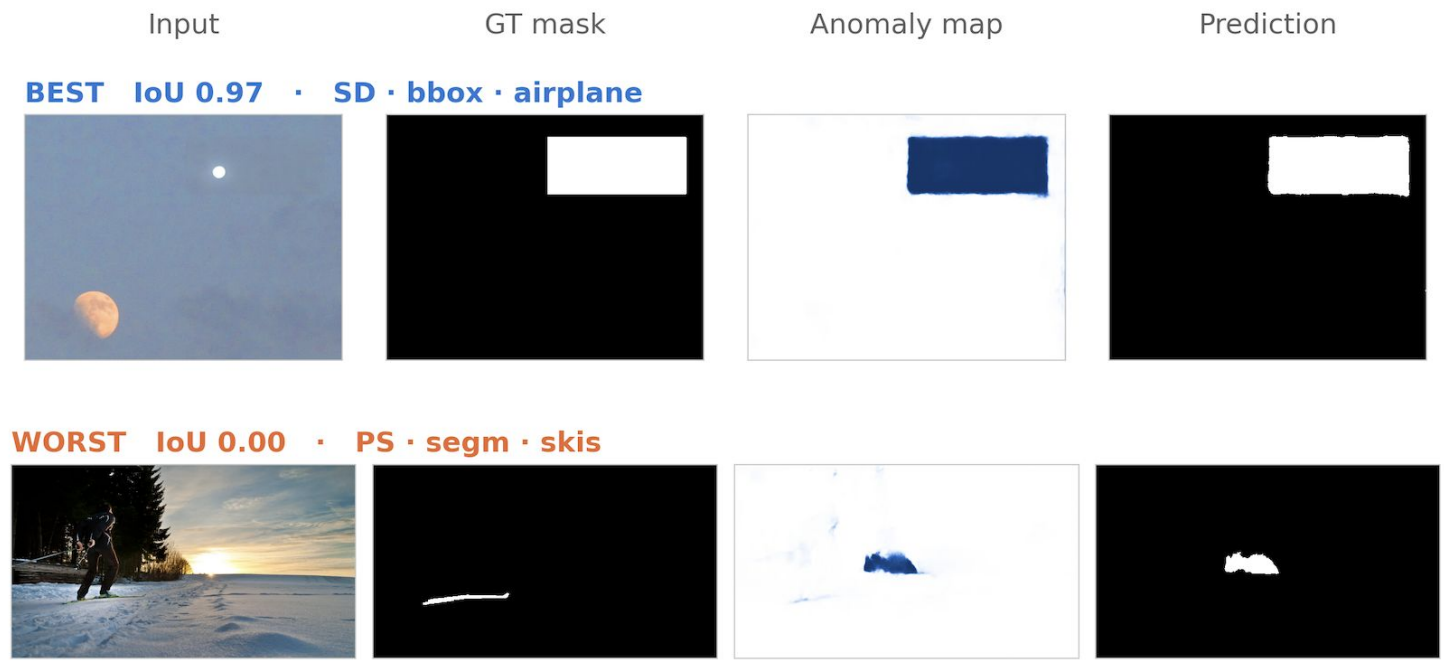
Detection with 1% threshold

TruFor: Transformer-Based Forgery Detection and Localization

- TruFor (CVPR 2023): **one image in, two answers out.**
- Image content + invisible noise "fingerprint" (Noiseprint++), combined by a **SegFormer-based transformer.**
- Outputs: integrity score (fake or not) and anomaly map (where).
- Official pretrained weights, no training on TGIF.



Localization: when TruFor works and when it fails



Detection: manipulated if score > 0.5 · Localization: map > 0.5 vs GT mask ≥ 128 · TGIF test set

Detection: what TruFor gets right, and where it fails

Input

Anomaly map

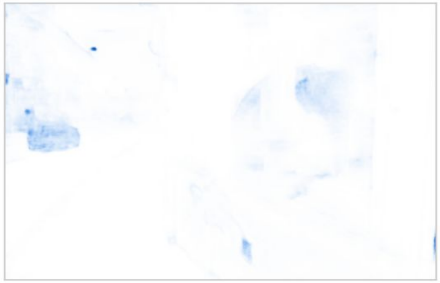
BEST: fake caught (SD): score > 0.999 → manipulated





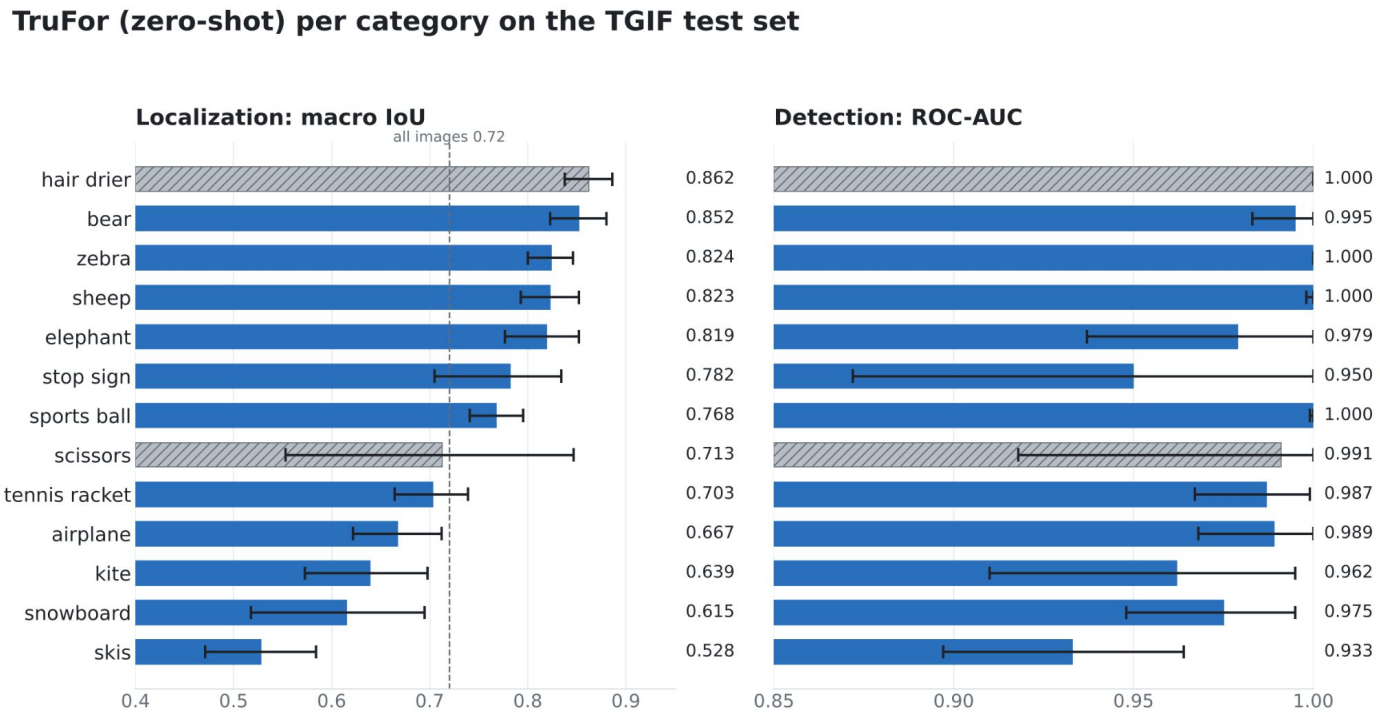
WORST: fake missed (SD): score 0.016 → authentic





Detection: manipulated if score > 0.5 · Localization: map > 0.5 vs GT mask ≥ 128 · TGIF test set

TruFor results by category



Bars: value over the 4,407 unique test images. Lines: 95% CI (bootstrap over source photos). Hatched grey = too few photos to rank (hair drier 2, scissors 8).
Note: bars start at 0.4 (IoU) and 0.85 (AUC), to make differences visible.

Challenges : reproducibility

Frequency-domain model:

- **ObjectFormer: no released weights and no frequency branch in the code**, so we had to train ObjectFormer ourselves, and trained only on the test set
- **CAT-Net** — all 6 files in the authors' Drive return *"the owner hasn't given you permission"*, despite a CC-BY licence
- the frequency-domain family has **no model we could evaluate as published**

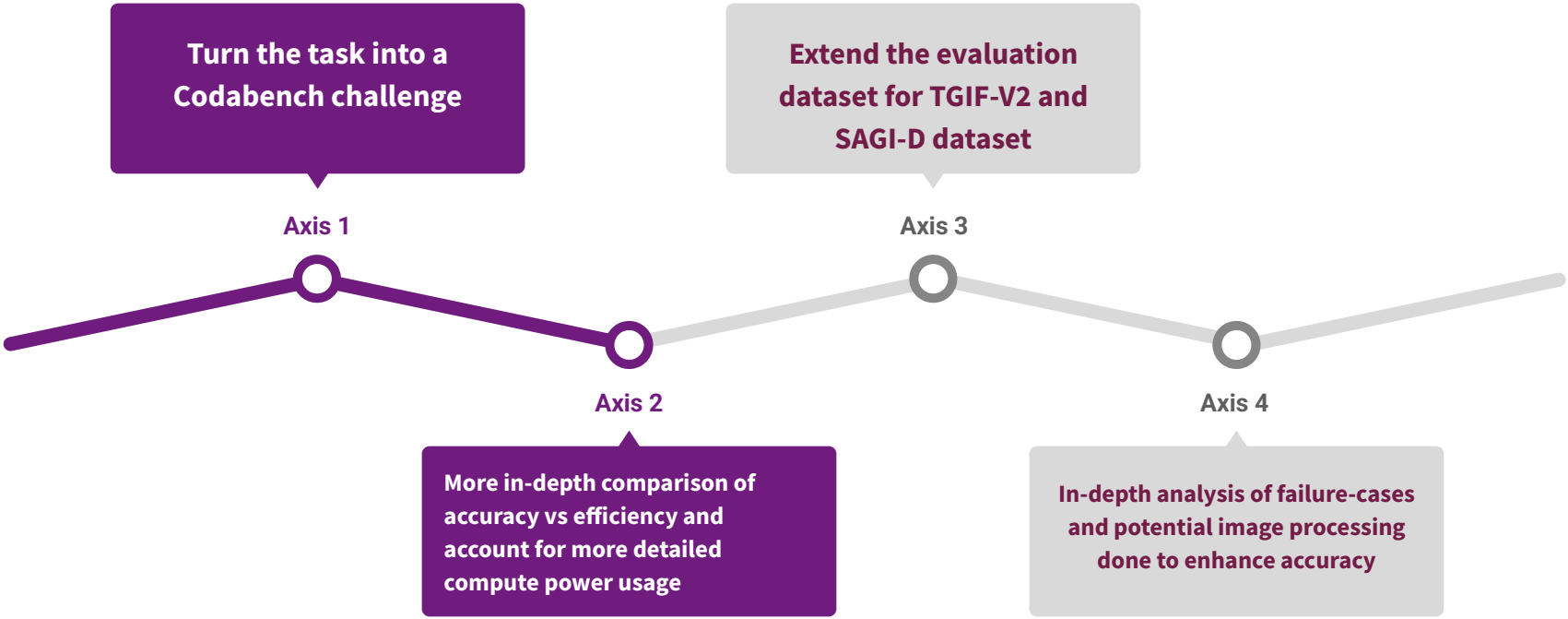
Limitations of the study

- **One dataset:** TGIF only, two inpainting sources (SD2, Photoshop). No cross-dataset generalization test, so everything we report is in-domain.
- **Limited and uneven categories:** 13 categories, very unbalanced — 38 source images for tennis racket, 7 for scissors, 2 for hair drier.
- **Limited compute power:** we had no GPU cluster and limited disk, so every model was trained or run on a reduced budget

Lessons learned

- Accuracy is not enough on an imbalanced set (92% fake)
- Detection ≠ localization
- Edit size and generator type strongly affect results
- Different categories and image nature affect the capacity of the detector
- A unified protocol is what makes models comparable
- Performance must be read together with model size and cost
- Various techniques exist for inpainting and it has many usages such as the technique of steganography

Perspectives



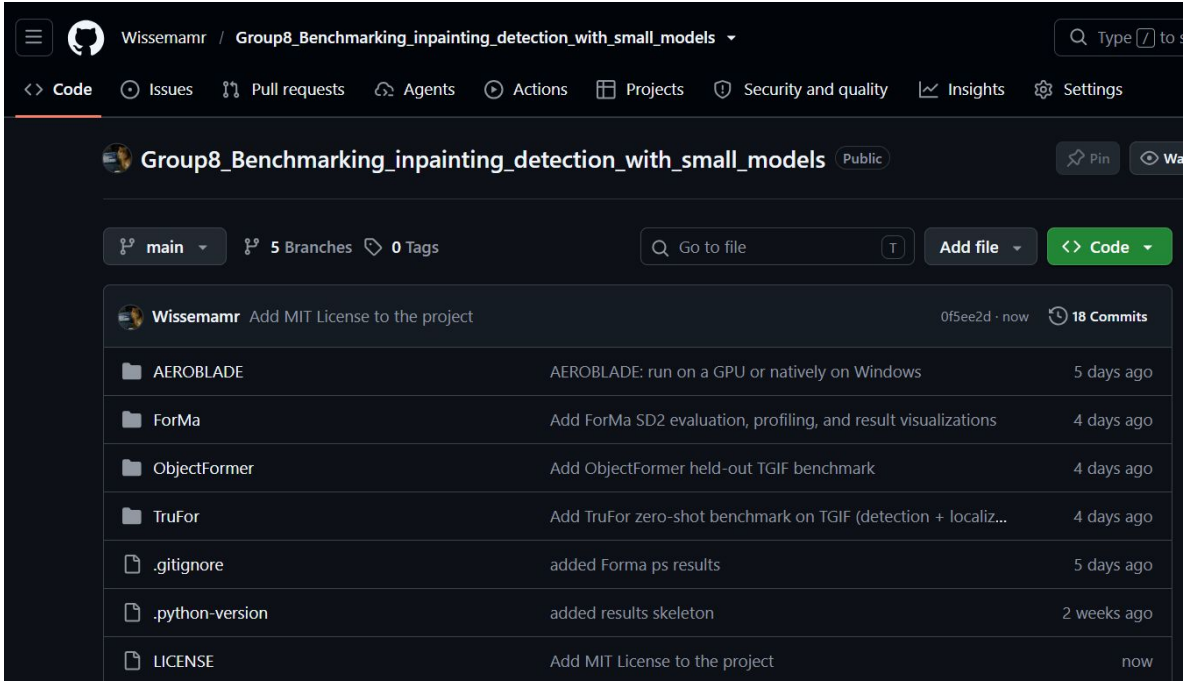
Our Team's Main Contributions



Hasti Hosseinpour	TruFor on the 3 resamples, detection + localization; per-category analysis and best/worst examples; state-of-the-art and model-categories slides
Wissem Amar	Experimenting with ForMa as an SSM model for localization and detection and reporting results, Repository organization
Dominika Chojnacka	PSCC-Net on the 3 resamples, on detection and localization, interpretation of the results, State-of-the Art of CNN-based methods, and datasets
Youssef El Otmani	AEROBLADE on the 3 resamples, full run, and on initial dataset, implementation of localisation task, analysing and interpreting results, computational comparaison run on all models, diffusion models state of the art research,
Sathya Gnanou	Frequency-domain model: trained ObjectFormer on the 3 resamples and audited its released code and weights, then tried CAT-Net.

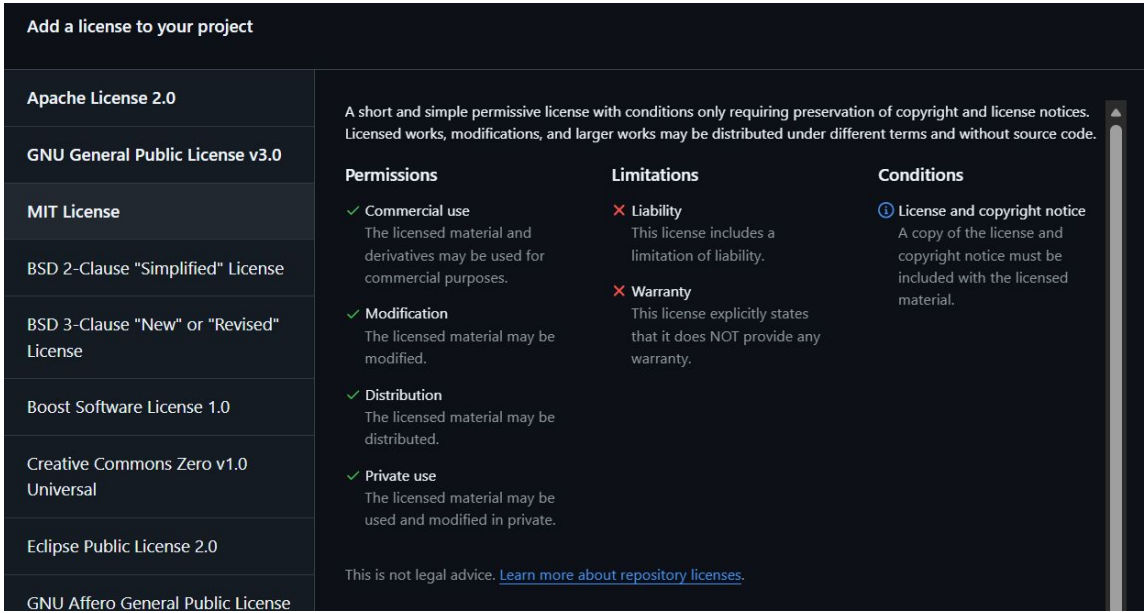
Documented Experiments Code

- The code is well documented, up-to-date and well organized
- One Folder per model experiments



Open Collaboration

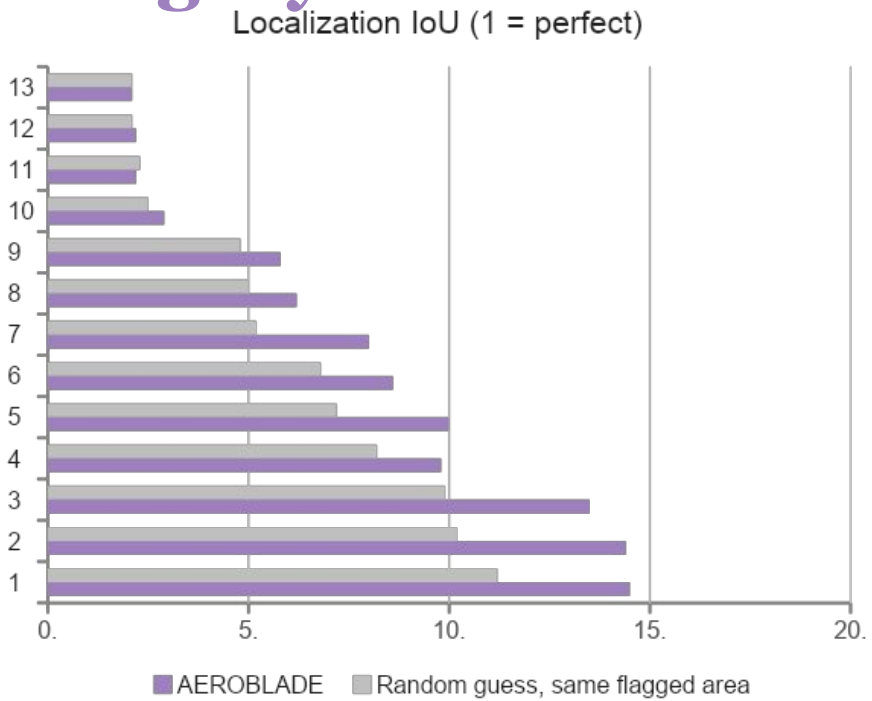
- We released our code under the MIT License to make our benchmark easy to reuse, adapt, and extend
- The goal is to encourage collaboration and move towards a community leaderboard

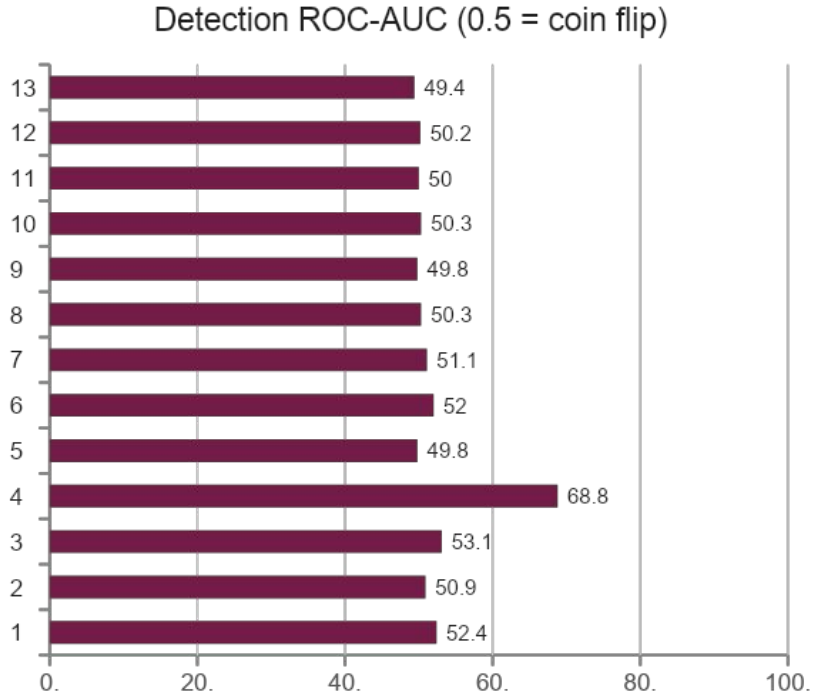


References

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Appendix: AEROBLADE results by object category

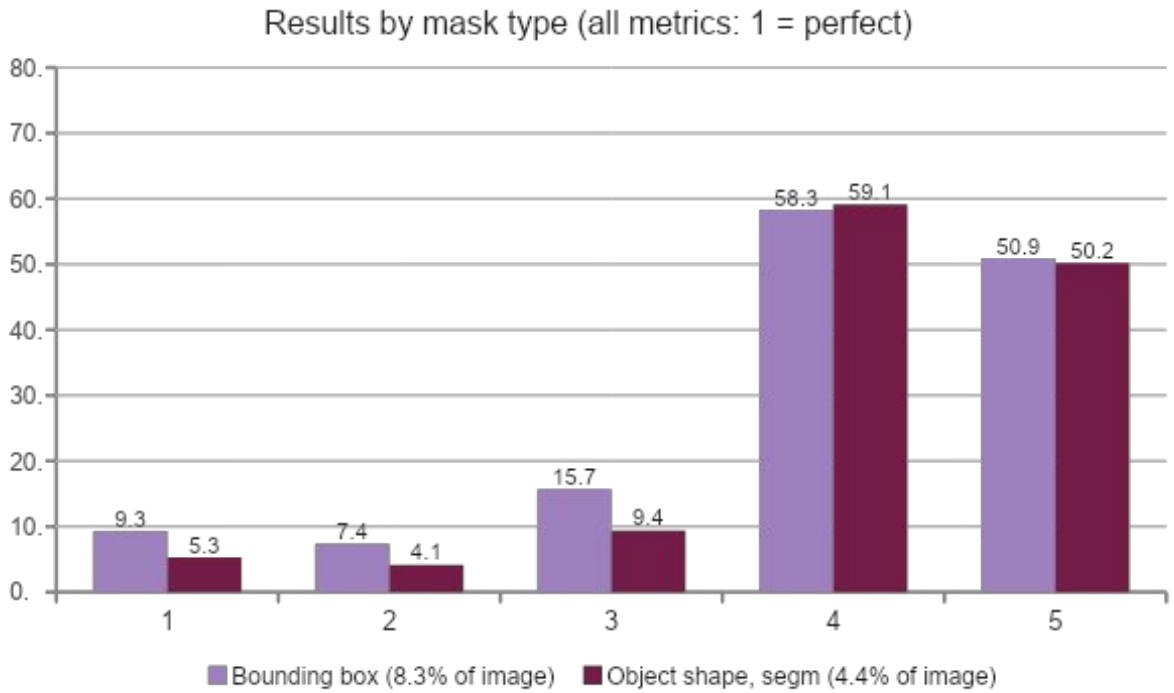




IoU follows object size, barely above a random guess · detection at chance in every category

(%) = average inpainted area. Detection: inpainted vs authentic photos of the same category; hair drier has only 2 authentic photos.

Appendix: AEROBLADE results by mask type



8.3% vs 4.4%

of the image inpainted: bbox vs segm

1.26× / 1.32×

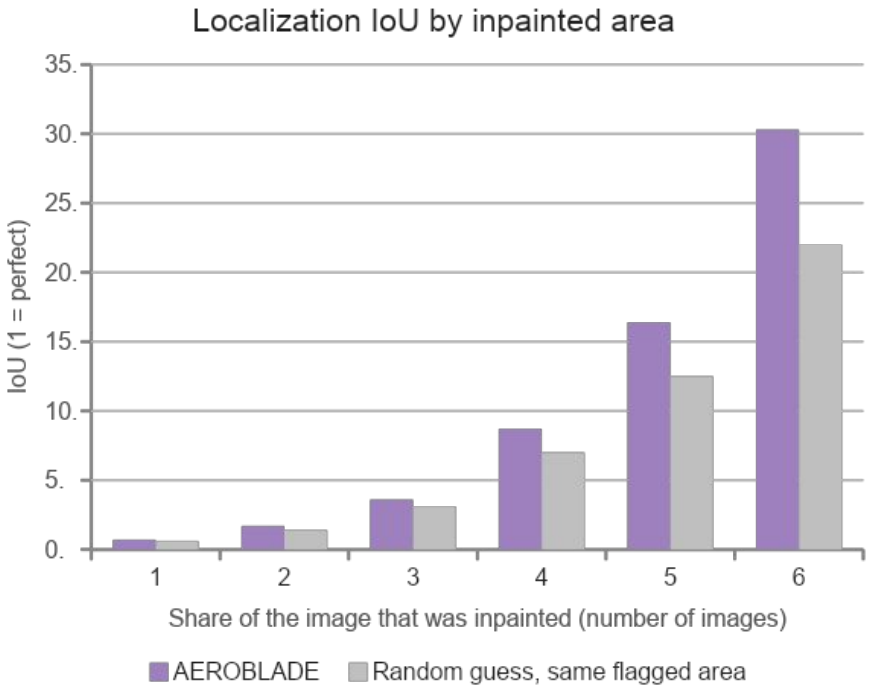
IoU over random guess: bbox / segm

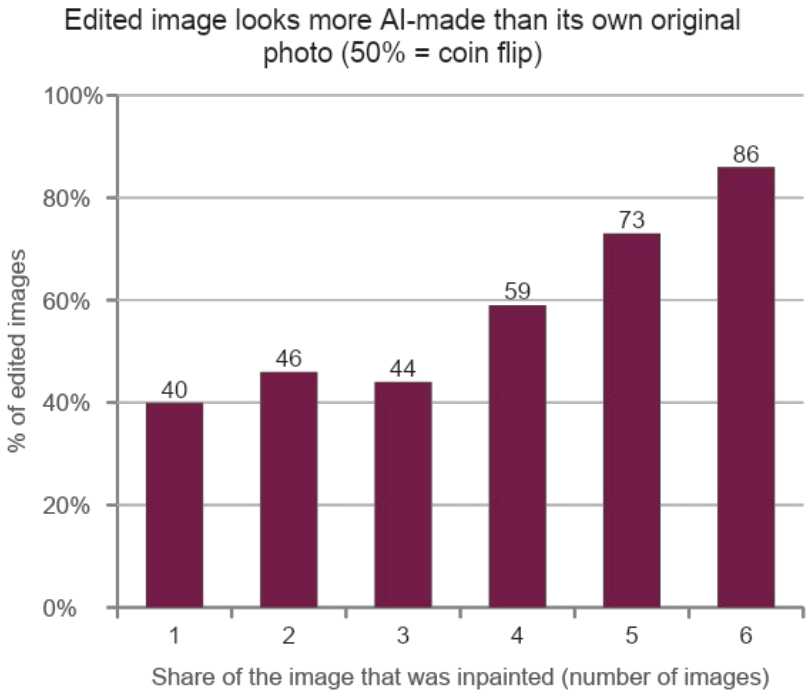
Boxes score higher only because they are bigger

Same behaviour for both mask types once the area is taken into account

Random guess: expected IoU when the same share of pixels is flagged at random. Detection: inpainted images vs all 343 authentic photos.

Appendix: AEROBLADE results by inpainted area

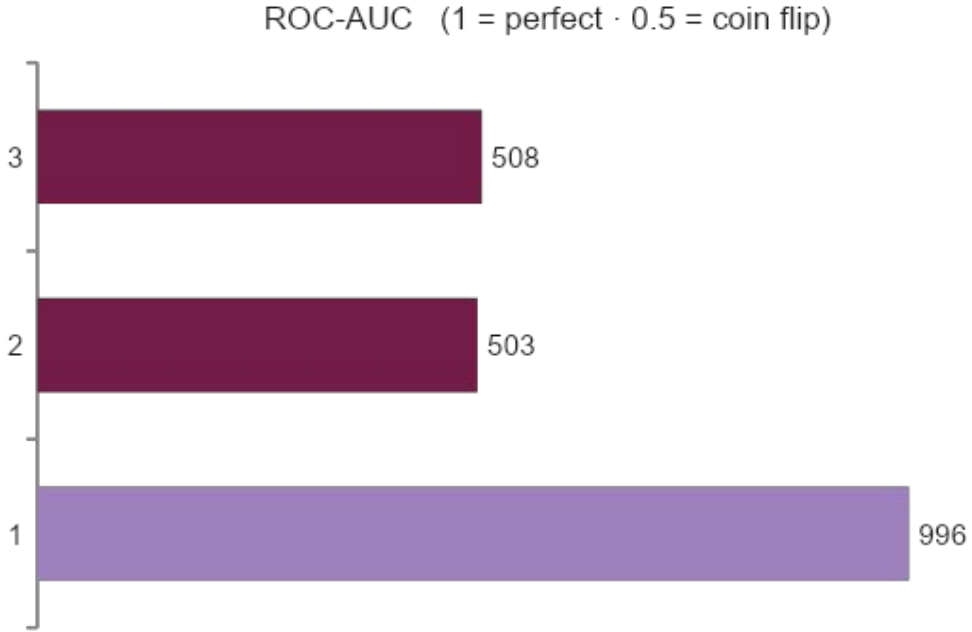




Big edits are noticed (86% for > 20%), but a 4% score shift is small next to the 47% spread between photos

(n) = number of inpainted images. Right: each inpainted image compared with its own untouched original; spread = interquartile range of authentic scores.

Detection: AEROBLADE cannot tell them apart



Typical score (median)

0.039	Authentic
0.039	SD2 inpainted
0.039	Photoshop inpainted

Same score → no threshold separates them

- Built for fully generated images: works
- Inpainting changes ~6% of the image: coin flip